C:\Users\dhruv\AppData\Loca

```
===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 1
Enter distribution point name: reservoir
Point added with ID: 0

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 1
Enter distribution point name: north colony
Point added with ID: 1

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 1
Enter distribution point name: south colony
Point added with ID: 2

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 1
Enter distribution point name: hospital zone
Point added with ID: 3
```

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6. Exit
Enter your choice: 1
Enter distribution point name: hospital zone
Point added with ID: 3

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 2

--- Distribution Points ---
0 : reservoir
1 : north colony
2 : south colony
3 : hospital zone
Enter first point ID: 0
Enter second point ID: 1
Enter installation cost: 4
Pipeline connection added successfully!

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 2

--- Distribution Points ---
0 : reservoir
1 : north colony
2 : south colony
3 : hospital zone
Enter first point ID: 0
Enter second point ID: 2
Enter installation cost: 3
Pipeline connection added successfully!

===== WATER PIPELINE NETWORK =====
```

```
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Enter installation cost: 3
Pipeline connection added successfully!

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 2

--- Distribution Points ---
0 : reservoir
1 : north colony
2 : south colony
3 : hospital zone
Enter first point ID: 1
Enter second point ID: 2
Enter installation cost: 2
Pipeline connection added successfully!

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 2

--- Distribution Points ---
0 : reservoir
1 : north colony
2 : south colony
3 : hospital zone
Enter first point ID: 1
Enter second point ID: 3
Enter installation cost: 5
Pipeline connection added successfully!

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
```

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3 : hospital zone
Enter first point ID: 1
Enter second point ID: 3
Enter installation cost: 5
Pipeline connection added successfully!

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 3

--- Distribution Points ---
0 : reservoir
1 : north colony
2 : south colony
3 : hospital zone

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 4

--- Available Pipeline Connections ---
From                 To                   Cost
reservoir            north colony         4
reservoir            south colony         3
north colony         south colony         2
north colony         hospital zone        5

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
```

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```
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 4

--- Available Pipeline Connections ---
From                 To                   Cost
reservoir            north colony         4
reservoir            south colony         3
north colony         south colony         2
north colony         hospital zone        5

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 5

--- Minimum-Cost Pipeline Network (Kruskal) ---
From                 To                   Cost
north colony         south colony         2
reservoir            south colony         3
north colony         hospital zone        5

All distribution points are connected.
Total minimum installation cost: 10
Pipeline connections selected: 3

===== WATER PIPELINE NETWORK =====
1. Add Distribution Point
2. Add Pipeline Connection
3. Display Distribution Points
4. Display Available Pipelines
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 6
Program terminated.

--------------------------------
Process exited after 283 seconds with return value 0
Press any key to continue . . .
```